Root Cause Analysis Report

Part	MS-4505-001 Sheet Metal Bracket
Material / process	SPCC Steel / bending
Case	rca_b593dfec9ac84436a7a4f0bad97e84a5
Date	2026-11-02 15:26:10
Status	AI proposal, pending expert review

1. Problem

Defect	Severity	Confidence	Detail
crack	critical	92%	
crack	critical	87%	
crack	critical	86%	
crack	critical	86%	
dent	major	54%	
scratch	minor	36%	
dent	major	27%	
crack	critical	22%	
dent	major	21%	
dent	major	18%	
dent	major	12%	

2. Root cause

Cause: Material Hardness Variation

Reason: The critical cracks detected on the SPCC steel bracket are directly linked to material hardness variation, which increases susceptibility to cracking during bending operations as described in [02cf0928-6739-4f7e-ae1b-f6740283f00f].

Effect: Cracking at bend radius

Confidence: 76%

Supporting evidence
- **02cf0928-6739-4f7e-ae1b-f6740283f00f**: Material Inconsistency / Hardness Variation Root Cause Analysis Defect Types: crack, deformation, dimensional variation Material: SPCC steel, cold-rolled sheet metal Root Cause: Hardness variation in incoming material lot causes unpredictable springback behavior and susceptibility to cracking under forming operations. Effect: Cracking at bend radius, dimensional inaccuracy due to variable springba

3. Solution

Corrective actions (fix this occurrence)

1. Verify material certificates against specification for yield and tensile strength
2. Conduct incoming material hardness testing

Preventive actions (stop it recurring)

1. Implement stricter incoming material quality control protocols
2. Audit material suppliers for consistency

Precautions

- Verify material certificates against specification for yield and tensile strength
- Conduct incoming material hardness testing
- Implement stricter incoming material quality control protocols
- Audit material suppliers for consistency

4. Other causes considered

Rank	Cause	Confidence	Reason
2	Tooling Damage or Die Wear	71%	The presence of dents and scratches alongside the cracks suggests potential die surface damage or tool wear, which is known to cause surface defects such as dents and scratches during the bending process [587aa665-c6f1-48df-8783-09d65ff4f82a].
3	Excessive Springback	35%	While the primary defects are cracks and dents, springback is a common root cause in sheet metal bending that can lead to stress concentrations, potentially exacerbating cracking issues if the bend radius is not properly compensated [6872e1ff-9d0f-41f8-9cd9-c783d66c1b46].

5. Traceability

AI model: gemini:gemini-3.1-flash-lite. Expert edited the cause: no.
This analysis was drafted with AI assistance from the retrieved knowledge-base evidence. It must be reviewed and confirmed by a qualified quality expert before corrective action is taken.